

DETAILED DESCRIPTION OF THE SYSTEM

The following detailed description is provided to enable a person skilled in the art to make and use the disclosed embodiments. The description is not intended to limit the scope of the disclosure. Variations, substitutions, and equivalents are contemplated.

A. System Components

In various embodiments, the system comprises a plurality of sensing nodes and a computing system. The sensing nodes may include one or more external nodes, one or more intraluminal nodes, and one or more context nodes. Each node may include a housing, one or more sensors, one or more processors or controllers, a power subsystem, and a communications subsystem.

The computing system may include one or more processors and memory configured to receive data from one or more nodes, store measurements, perform signal processing, derive features and metrics, perform correlation and analysis across nodes and time windows, and present privacy-preserving outputs. The computing system may be implemented on a mobile device, tablet, desktop computer, server, cloud platform, or combinations thereof.

B. Data Collection Modes

Nodes may operate in continuous mode, periodic sampling mode, event-triggered mode, or session-based mode. In continuous mode, one or more sensors may sample at a sustained rate across extended wear. In periodic mode, sensors may sample at scheduled intervals, adaptively based on motion state, or in response to detected events. In session-based mode, measurements may be collected during a guided routine such as baseline acquisition, geometry scanning, compliance mapping, or contact validation.

C. Time Windows and Longitudinal Tracking

In certain embodiments, the computing system segments measurements into time windows including fixed windows, rolling windows, and event windows. The computing system may compute metrics within each window and generate longitudinal views across multiple windows. Longitudinal outputs may include trend curves, variability summaries, stability measures, and confidence indicators, enabling interpretation across nights, days, weeks, or longer periods.

D. Privacy-Preserving Representation

In various embodiments, the system presents results without explicit anatomical imagery and without explicit anatomical labeling. Outputs may be represented using abstract indices, zones, ranges, and curves. In certain embodiments, outputs are normalized relative to a user's baseline, a user's historical distribution, or a population-referenced model, while preserving privacy and reducing stigma.

E. Cross-Device Correlation and Contextual Interpretation

In certain embodiments, the computing system correlates outputs derived from different nodes, including by aligning timestamps, detecting co-variation, computing lag relationships, and computing correlation confidence. Context signals from an upper-body node may be used to condition interpretation of intimate-region patterns, including distinguishing physiologic changes from motion artifacts or environmental changes. Baseline acquisition results from a session-based node may be used to normalize

and interpret longitudinal patterns collected by wearable nodes.

F. Cross-User Correlation

In certain embodiments, the system supports privacy-preserving correlation between metrics of a first user and metrics of a second user. The correlation may be computed using abstracted metrics, time-window summaries, stability measures, and/or timing synchrony measures. Cross-user correlation may be supported across a range of pairing contexts including male-to-female, male-to-male, female-to-female, and other pairings, without requiring explicit anatomical imagery or explicit anatomical labeling.

G. Transition to Node-Specific Embodiments

The sections below describe representative node embodiments. Each node description includes example form factors, sensor suites, sampling approaches, wear validation, artifact rejection, and example outputs. Any node may be used independently, and any node may be used in combination with one or more other nodes to enable cross-device correlation and longitudinal interpretation.